

Functionalities

Before programming further, we need to create the various functionalities of the player, the allies, and the enemies. The reaction and input to these functionalities are from the player input or through AI (artificial intelligence). Unreal has gameplay classes for such functionalities such as:

1. **GameMode** – This covers the game rules and win conditions. It has mostly static data that are discreet to the user. It has two states and exists only on the server.
2. **GameState** – It is the state of the game like connected players, scores, list of missions completed. It is available on the server and all the clients are free to update the data.
3. **PlayerState** – It denotes the state of participation of the player. It exists on all machines.

All these classes aid us in creating characters that can be created for the game.

Creating a custom GameMode

1. In the Content Browser, right-click and select Folder.
2. Name the folder as Blueprints.
3. Double-click the Blueprints folder to open into it.
4. In the Content Browser, right-click and select Blueprint Class.
5. In the Pick Parent Class window that appears, select Game Mode.
6. Name your new Game Mode `ArcadeShooter_GameMode`.
7. Select File > Save All (or press Ctrl+S).

While the default game mode is available in:

1. Select Edit > Project Settings.
2. In the Project Settings panel, click the Maps & Modes section.
3. In the Default Modes section, click the Default GameMode field to open the search box for all Game Modes.
4. Select your newly created `ArcadeShooter_GameMode` Game Mode.

Players

Adventure games require a player, an enemy, a few spectators, and some helping characters to keep the fire burning. Story-based or episodic adventures may need a narrator too. In this section we'll take a look at how we can develop various characters to be used in the game. The Unreal environment has two types of player entities named as: